

TAREK RAHMAN

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in iamtarekrahman | 🌐 GitHub | 📁 GrabCAD 📁 Portfolio

EDUCATION

IUBAT

(International University of Business
Agriculture and Technology)
B.SC. - MECHANICAL ENGINEERING
May 2021 - June 2025 | Dhaka
Cum. GPA: 3.90/4.0

Thesis title: Experimental and
Computational Analysis of a Tesla
Turbine: Impact of Water Velocity, Disk
Gap, and Disk Thickness
Supervisor: Dr. Debasish Sarker

ITSC

(Ibn Taimiya School and College)
H.S.C. - SCIENCE
July 2017 - July 2019 | Cumilla
GPA: 4.50/5.0

CERTIFICATIONS

Risk Management and Industrial Safety
Credential ID: XYKIEIMDLFIA

Renewable Energy
Credential ID: EZSSNCP2PM3U

Autodesk Certified Professional:
AutoCAD for Design and Drafting Prep
Credential ID: FOE92N2S5Y9C

Certified SolidWorks Associate (CSWA)
Credential ID: C-7CWAKR3PD6

ORGANIZATIONS

Member of ASME
Member of IMechE

SKILLS

SOFTWARE

SolidWorks • AutoCAD • MS Office
(Advance Excel, PowerPoint, Word)

PROGRAMMING

C (Fundamental) • LaTeX

DATA ANALYSIS

Data Analysis Reporting • Technical
Documentation • Bill of Materials
(BOM) • Vendor & Cross-functional
Coordination

DESIGN & ENGINEERING

Mechanical Design • Manufacturing
Processes • GD&T • Tolerance Analysis
• Process Optimization • Feasibility
Study Preparation

LANGUAGES

Bengali (Native)
English (Fluent)

SUMMARY

Mechanical Engineer with CSWA certification and experience in cross-functional project coordination, vendor management, and technical documentation. Proven ability to design, simulate (ANSYS/SolidWorks), and fabricate mechanical systems. Eager to leverage hands-on prototyping and design skills in an R&D environment.

WORK EXPERIENCE

HUNAN CONSTRUCTION & ENGINEERING GROUP (HCEG)

ASSISTANT MECHANICAL ENGINEER

Jan 2026 – Present | Rajshahi WASA Surface Water Treatment Plant Project

- Coordinate with equipment manufacturers, consultants, and senior engineers to prepare, revise, and track technical/material submittals (pumps, valves, crane etc.) for a large infrastructure project.
- Review vendor documentation against Employer's Requirements and international standards (BS, ISO, IEC), resolving consultant non-compliance comments, verifying spare parts/performance data, and maintaining accurate records across resubmission cycles

HEIDELBERG MATERIALS BANGLADESH PLC.

INTERNSHIP- MAINTENANCE AND OPERATIONS

Feb 2025 – May 2025 | Mukterpur Plant

- Rotated through maintenance, production, and QC to gain operational knowledge using SAP, while analyzing data to prepare performance reports, identify inefficiencies, and enhance equipment reliability through cross-functional collaboration.

PROJECTS

TESLA TURBINE

- Designed and fabricated a Tesla turbine using machine shop tools (lathe, milling), evaluating torque, power, and efficiency under varying conditions, focusing on velocity, disk gap, and thickness.
- Performed CFD analysis in ANSYS to study flow behavior within the Tesla Turbine, validate design performance, and identify parameters influencing torque, efficiency, and velocity distribution.

JOURNAL BEARING

- Developed a Journal Bearing design in SolidWorks for a Machine Design course to evaluate performance under varying load and speed conditions, selecting the most suitable configuration based on theoretical and simulation analyses.

SCHOLARSHIPS & AWARDS

- Ahsanullah Scholarship - Merit-based award for outstanding academic performance (Summer 2023)- IUBAT
- Award for Academic Excellence - Recognized for top academic achievement (Fall 2022, and Fall 2023)- IUBAT

EXTRACURRICULAR ACTIVITIES

- Researcher
NARP - Nayem's Association for Research & Publication | 2026 - Present
- Co-media and Communication Secretary
ISME - IUBAT Society of Mechanical Engineers | 2022 -2023